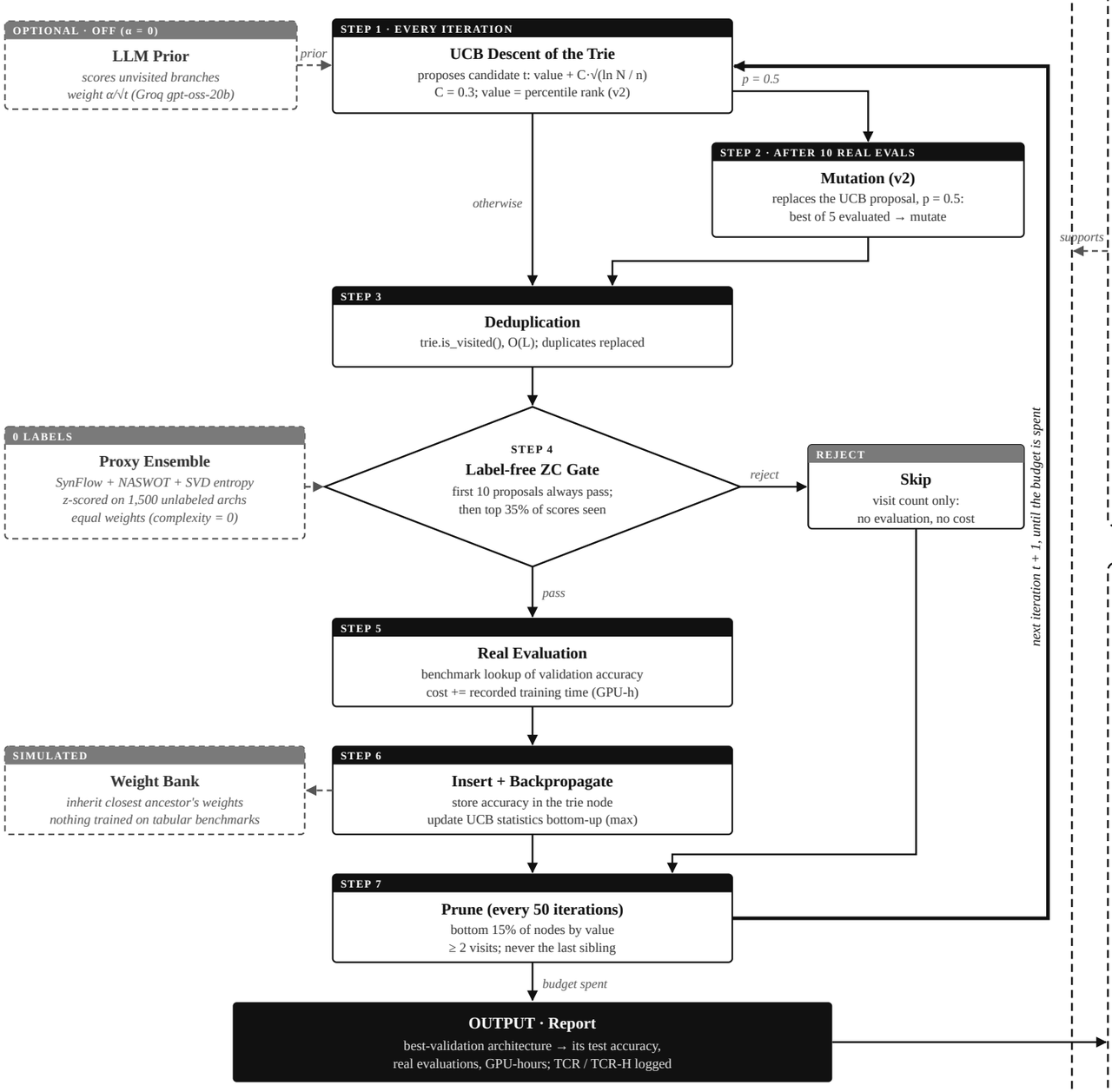
ArchTree v2 — Search Pipeline, Explained

One search run per (dataset, seed, budget). Left: the loop as implemented in archtree/search.py. Middle: what supports and evaluates it. Right: each block in plain terms.

CORE SEARCH LOOP



INFRASTRUCTURE

supports

Groq Key Pool + Rate Limiter
round-robin keys, RPM / TPM pacing

Shared Caches
LLM answers and proxy scores

Storage Layer
atomic JSON writes; CSV + logs

Checkpoint / Resume
finished runs skipped on rerun

Parallel Workers
proxy pre-compute, equal GPU-h

EVALUATION LAYER

Baselines
9 external + internal arms

Equal GPU-Hours Study
fair compute, 13 methods, 3 benchmarks

Ablation Suite
A1–A7, design, proxy, UCB C

Statistics
Wilcoxon + Holm, Friedman / Nemenyi,
TOST, P(improvement), Cohen's d

CORE LOOP — IN PLAIN TERMS

1. **UCB proposal**
   Each iteration, walks the tree of design choices to propose one new architecture.
2. **Mutation**
   After 10 real tests, half the time swaps that proposal for a tweak of a top design.
- **LLM prior**
  An AI model guesses which branches look promising. Studied separately; off in the main system.
3. **Deduplication**
   The tree remembers every design, so no design is ever paid for twice.
4. **Zero-cost gate**
   A one-pass check on an untrained network; only the top 35% go on to a real test.
- **Proxy ensemble**
  Three training-free scores averaged with equal weight. Needs no accuracy labels.
5. **Real evaluation**
   Looks up the design's accuracy — the expensive step the pipeline tries to minimise.
6. **Backpropagate**
   Feeds the result back up the tree so future choices get smarter.
- **Weight bank**
  Lets a child reuse its parent's weights. Simulated here, since nothing is trained.
7. **Prune**
   Every 50 steps, cuts the weakest 15% of branches so the search stays focused.
- **Report**
  Picks the best design on validation data and reports its test accuracy and cost.
- **TCR / TCR-H**
  How much of the space was covered; TCR-H also rewards good accuracy–latency trade-offs.

INFRASTRUCTURE & EVALUATION

- [ ] **Groq key pool**
  Rotates several API keys so rate limits never stall the LLM studies.
- [ ] **Caches & storage**
  Every answer, score and result is saved safely, so nothing is recomputed or lost.
- [ ] **Checkpoint / resume**
  A stopped pipeline restarts where it left off.
- [ ] **Baselines & ablations**
  Compare against other methods, and switch one part off at a time to prove it helps.
- [ ] **Statistics**
  Tests if differences are real (Wilcoxon, Holm), ranks methods (Friedman) and checks equivalence (TOST).